Zixuan Liu

CONTACT

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RESEARCH INTEREST

My research focuses on developing intelligent surgical assistive systems that integrate mixed reality (MR), robotics, computer vision, and deformable registration to enhance the efficiency of minimally invasive surgeries. I am particularly interested in (1) utilizing MR and computer vision algorithms to provide intraoperative guidance and navigation, (2) developing robotic systems with intelligent decision strategies for understanding surgical scene, and (3) deformable modeling for soft tissue surgery, especially for breast-conserving surgery (BCS).

EDUCATION

Ph.D. in Computer Science 08/2024 – now
Vanderbilt University
Current GPA: 4.0/4.0
Affiliations:
Biomedical Modeling Laboratory (BML)
Machine Automation, Perception, and Learning (MAPLE) Lab
Vanderbilt Institute for Surgery and Engineering (VISE)
Primary advisor: Michael I. Miga; *Co-advisor:* Jie Ying Wu
Research Keywords: Computer Aided Intervention, Image-guided Surgery, Surgical Robotics, Mixed Reality

M.S.E. in Computer Science 08/2022 – 05/2024
Johns Hopkins University
GPA: 3.79/4.0
Affiliations:
Laboratory for Computational Sensing and Robotics (LCSR)

B.S. in Computer Science, *Magna Cum Laude* 08/2018 – 06/2022
minor in Mathematical Science
Wenzhou-Kean University
GPA: 3.818/4.0

SELECTED AWARDS

Selected Awards

7. Finalist, **Best Paper Award** 2024
For paper [J-3] at IPCAI 2024.

6. **Potential Impact Award**, Introduction to Augmented Reality 2023
Johns Hopkins University (Awarded by Prof. Alejandro Martinez Gomez)

5. **Best Project Award**, Computer Integrated Surgery II 2023
Johns Hopkins University (Awarded by Prof. Russell Taylor)

4. **Dean's Scholarship – Research and Innovation** 2021
Wenzhou-Kean University

3. **Dean's Scholarship – Second Class** 2021
Wenzhou-Kean University

2. **Dean's Scholarship – Third Class** 2020
Wenzhou-Kean University

1. *Magna Cum Laude*, B.S. in Computer Science 2022
Wenzhou-Kean University, *GPA: 3.818/4.0*

Professional Employment

Robot Algorithm Engineer Intern 07/2023 – 09/2023
Euclid Techlabs LLC, Hybrid in Beltsville, MD, USA
– Involved in development of HDR Electronic Brachytherapy system prototype.

Publications

I have (co)-authored 4 journal articles, 4 conference papers, and 1 book chapter, with 1 paper currently under review. A detailed list is available below and on Google Scholar.

Peer-reviewed Journal Articles

[J-4]. **Z. Liu**, B. Xiang, J. Heiselman, J.Y. Wu, I. Meszoely, M.I. Miga. "Robotic-enabled Surgical Navigation Assistant for Breast-Conserving Surgery." *IEEE Robotics and Automation Letters (RA-L)*, 2026.

[J-3]. B.D. Killeen, **Z. Liu**, H. Zhang, L. Wang, C. Kleinbeck, M. Rosen, R.H. Taylor, G. Osgood, M. Unberath. "Stand in Surgeon's Shoes: Virtual Reality Cross-training to Enhance Teamwork in Surgery." *International Journal of Computer Assisted Radiology and Surgery*, 2024.
Special Issue: *Information Processing in Computer-Assisted Interventions (IPCAI) 2024*
Finalist for **Best Paper Award** at IPCAI'24.

[J-2]. D. Liu, G. Li, S. Wang, **Z. Liu**, Y. Wang, L. Connolly, G. Shen, K. Cleary, I. Iordachita. "A Magnetic Resonance Conditional Robot for Lumbar Spinal Injection: Development and Preliminary Validation." *International Journal of Medical Robotics and Computer Assisted Surgery (IJMRCARS) 2023.*

[J-1]. K.E. Ehimwenma, S. Krishnamoorthy, **Z. Liu**, Y. Qiu, Y. Liu, W. Dou. "Optimal Recycle Price Game Theory Model for Second-hand Mobile Phone Recycling." *Environmental Science and Pollution Research International (ESPR)*, 2021: 1–16.

Peer-reviewed Conference Papers

[C-4]. B. Xiang, **Z. Liu**, M.I. Miga. "Robotic Stereo-camera System with SAM 2 for Multiview Surface Reconstruction of Anatomical Structures in Surgery." *Medical Imaging 2026: Image-Guided Procedures, Robotic Interventions, and Modeling*, SPIE, vol. 13927, p. 139270P, 2026.
Selected for oral presentation.

[C-3]. U. Rajamani, **Z. Liu**, I.M. Meszoely, M.I. Miga. "Real-time Optimal Viewpoint Control for Image-guided Surgery using Learning-based Active Perception." *Medical Imaging 2026: Image-Guided Procedures, Robotic Interventions, and Modeling*, SPIE, vol. 13927, p. 1392721, 2026.

[C-2]. Q. Yang, F. Li, J. Xu, **Z. Liu**, S. Sridhar, W. Jin, J. Du, J. Heiselman, M. Miga, M. Topf, J.Y. Wu. "Augmented Reality-Based Guidance with Deformable Registration in Head and Neck Tumor Resection." *International Conference on Medical Image Computing and Computer-Assisted Intervention (MICCAI)*, 2025.

[C-1]. D. Liu, **Z. Liu**, G. Li, L. Connolly, G. Shen, I. Iordachita. "Development of an MRI Guided Auxiliary Robot for Spinal Injections." *IEEE International Conference on Robotic Computing (IRC)*, 2023.

Book Chapters

[B-1]. Y. Liu, W. Dou, **Z. Liu**, E. Szarek, S. Krishnamoorthy. "Integration of IoT and Big Data in the Field of Entertainment for Recommendation System." In *Securing IoT and Big Data*, pp. 109–121. CRC Press, 2020.

Preprints / Under Review

[P-1]. M. Liu, **Z. Liu**, R. Cai, Y.-C. Ku, S. Gu, A. Jain, A. Martin-Gomez, M. Armand. "X-GuideAR: An Augmented Reality Framework to Mitigate Radiation Exposure during Fluoroscopic Guidance. A Use Case in S2AI Screw Placement." *[Under Review]*

Teaching

Peer Tutor 2019 - 2021
Student Academic Support Service Center
Wenzhou-Kean University

Talks and Press

Invited Talks and Demos

[P-4]. 2026 International Symposium on Medical Robotics (ISMR) 04/2026
Late-Breaking Research - Poster Presentation
Knoxville, USA
"ReSNA: A Robotic-Enabled Surgical Navigation Assistant for Breast-Conserving Surgery."

[P-3]. 14th VISE Symposium 12/2025
Oral Presentation
Vanderbilt University, USA
"ReSNA: A Robotic-enabled Surgical Navigation Assistant for Breast-Conserving Surgery with Deformable Correction and Augmented Reaility."

[P-2]. 13th VISE Symposium 12/2024
Vanderbilt University, USA
"Augmented Reality Robotic-enabled Surgical Navigation Assistant for Breast-Conserving Surgery"

[P-1]. 1st Computer Science Symposium 12/2024
Wenzhou Kean University, China
"Optimal Recycle Price Game Theory Model for Second-hand Mobile Phone Recycling"

Selected Press

[P-1]. Our work [J-3] was featured in a JHU Computer Science News article on the IPCAI 2024 conference.

Service

Computer Integrated Surgery I (EN.601.455/655)
Department of Computer Science, Johns Hopkins University
Course Assistant Fall 2023

MSE Computer Science Admissions Committee
Department of Computer Science, Johns Hopkins University
Student Reviewer Fall 2023, Fall 2024

Journal Reviewer
- IEEE Transactions on Medical Robotics and Bionics